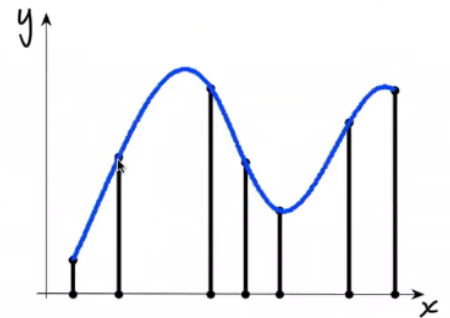
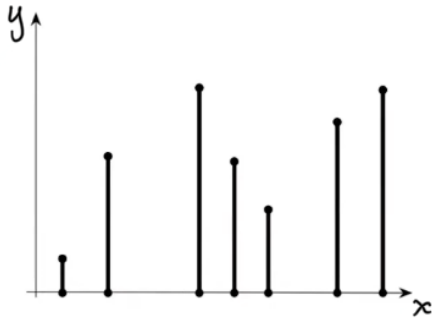
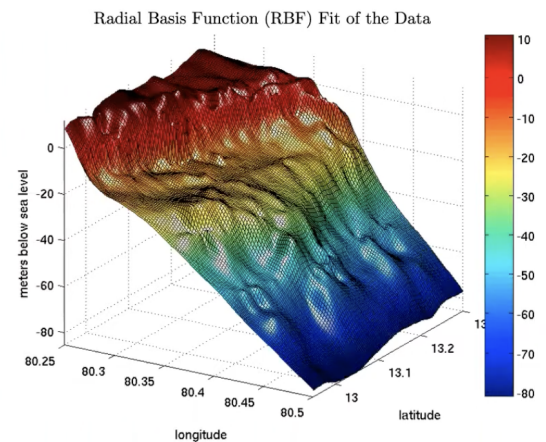
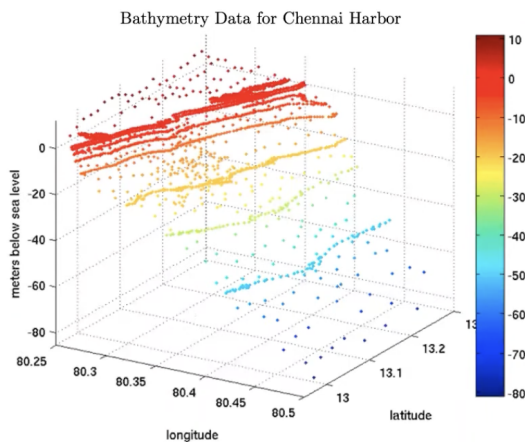


MAT 425, Class 10: Radial Basis Function (RBF) Interpolation (*Section 2*)

So far, we have focused on interpolation with 1D data, reconstructing a continuous curve that passes through each given data point. This process ensures that the interpolated function accurately represents the underlying trend of the data while maintaining smoothness and continuity.



Now, we extend interpolation to higher dimensions. For example, with 2D data, we aim to construct a continuous surface that interpolates a given set of points. This approach generalizes further to even higher dimensions, where we interpolate functions over multi-dimensional spaces while maintaining continuity and smoothness.



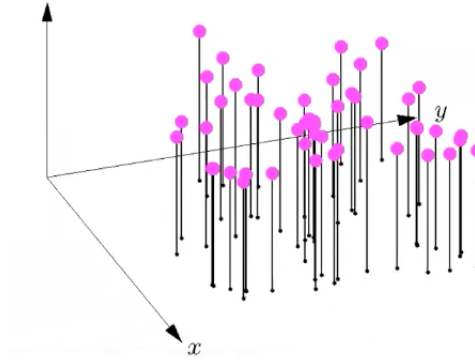
Here, we have bathymetry data for Chennai Harbor, where longitude and latitude serve as the independent variables, and depth (meters below sea level) is the dependent variable. By constructing a continuous surface that interpolates this data, we obtain a smooth representation of the underwater topography. This interpolated surface can then be used in computational models to simulate tsunami propagation, aiding in risk assessment and disaster preparedness.

To model real-world phenomena like tsunami simulations, we need **gridded data** or a **continuous representation** of topography. However, standard 1D interpolation methods do not generalize well to higher dimensions, making it difficult to construct a smooth interpolating surface.

This is where **meshfree methods** excel. Unlike traditional grid-based approaches, meshfree methods provide flexibility in handling scattered data without requiring structured grids. One powerful meshfree approach is **Radial Basis Function (RBF) interpolation**. This method can be generalized to any dimension, is simple to program and use, and works well with scattered data.

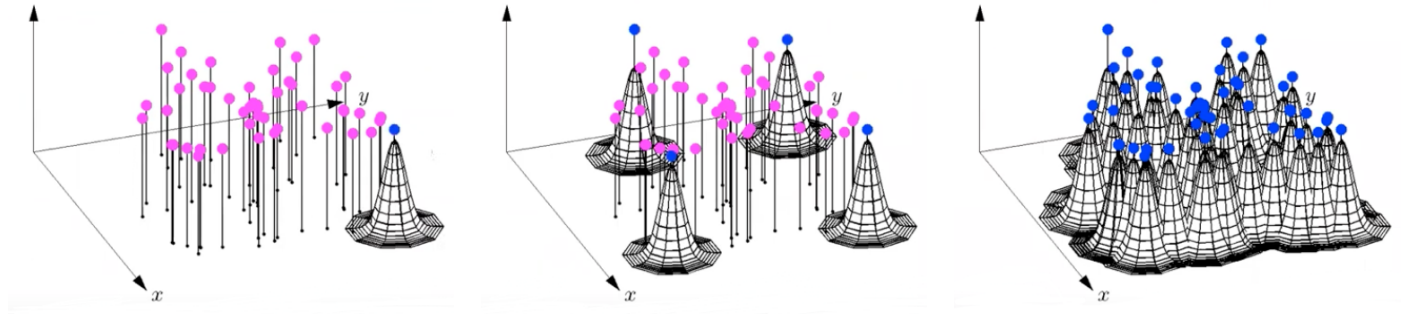
Now, let's explore the key idea behind **RBF interpolation**.

Suppose we are given N data points $(\mathbf{x}_j, f_j)$ for $\mathbf{x}_j \in \Omega \subset \mathbb{R}^2$, where $j = 1, \dots, N$. Our goal is to find an interpolant $P(\mathbf{x})$ satisfying $P(\mathbf{x}_j) = f_j$ for each j .

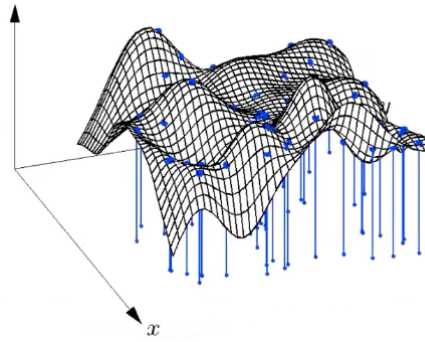


Basic RBF interpolant for $\Omega \subset \mathbb{R}^2$

Geometrically, the idea is as follows: We take the **kernel**, translate it to a point $\mathbf{x}_j$, and then rotate it around. We repeat this process for all N points. This approach allows us to build the interpolant by considering how the kernel behaves in relation to each data point and its position in the domain $\Omega \subset \mathbb{R}^2$.



Now, we want to **weight** these translated and rotated kernels and sum them in such a way that we obtain a surface passing through the given data points. This means we need to determine the coefficients c_j , which control the contribution of each kernel at the data points.

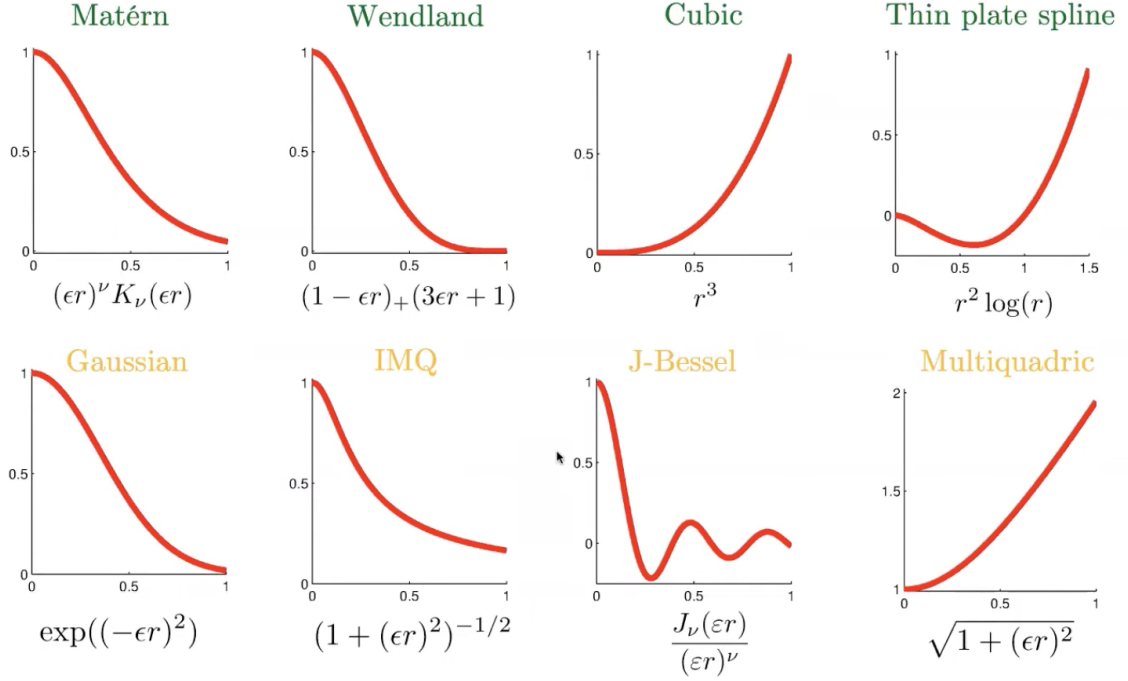


Mathematically, this translates to solving a **linear system of equations**, where we enforce that the function s must satisfy the interpolation condition: $P(\mathbf{x}_j) = f_j$, for $j = 1, \dots, N$. Thus, we construct a system ensuring that our interpolant exactly matches the given data values.

Depending on how we choose the function ϕ , we can guarantee that the matrix A is **invertible**, regardless of how the points $\mathbf{x}_j$ are distributed. This means that we do not require our data to lie on any specific grid, ensuring a **unique solution** to the problem.

The key property of this function is that it depends only on the **Euclidean distance**, i.e., the norm. Therefore, all we need to compute are distances between points, regardless of the dimensionality of the space we are working in.

Here are some examples of common **kernels**:



Radial basis functions (RBFs) can be broadly categorized into two groups. The first category consists of functions that are not infinitely differentiable. The second category includes functions that are infinitely differentiable.

The choice of radial function affects the reconstructed surface. Specifically, the smoothness of the reconstructed surface depends both on the **smoothness of the data** and the **smoothness of the chosen radial function**. In our **brain imaging experiment**, or rather in the paper we selected, authors use the **Gaussian kernel**.

Definition: Radial Basis Function

A function whose value depends only on the **magnitude** of its argument is called a **radial function**. Mathematically, this means:

$$\phi(\|\mathbf{x}\|) = \phi(r), \quad \text{where } \phi : [0, \infty) \rightarrow \mathbb{R}$$

where r is the length (norm) of the vector $\mathbf{x}$.

This property implies that ϕ is **constant for all input vectors of the same length**, which is why we refer to ϕ as a **radial basis function**.

Example 1. Suppose we have three data points: $x_1 = 1$, $x_2 = 3$, $x_3 = 3.5$ with corresponding function values: $f_1 = 1$, $f_2 = 2.2$, $f_3 = 0.1$. We choose the radial basis function: $\phi(r) = e^{-r^2}$. Find the RBF interpolant $P(x)$.

Example 2. We are given the following data points: $(x_1, y_1) = (0, 1)$, $(x_2, y_2) = (1, 3)$, $(x_3, y_3) = (2, 2)$. The Gaussian RBF function is defined as: $\phi(r) = e^{-\beta r^2}$, where $\beta = 1$. Find the RBF interpolant $P(\mathbf{x})$.

Reference

- Fasshauer, G. E. (2007). Meshfree approximation methods with MATLAB (with CD-ROM) (Vol. 6, Interdisciplinary Mathematical Sciences). World Scientific Publishing. ISBN: 978-981-270-633-1 (hardcover), 978-981-270-634-8 (softcover), 978-981-3101-57-9 (ebook)
- Wright, G. (2021, April 9). Meshfree methods for scientific computing [Video]. YouTube. Boise State University PhD in Computing. <https://www.youtube.com/watch?v=jmmDv7BxUnM>
- De Marchi, S., & Perracchione, E. (2018, January 24). Lectures on radial basis functions. Department of Mathematics “Tullio Levi-Civita,” University of Padua.